

# DB4BM loudspeaker range up to 25 watts

Hazardous & ordinary locations



## Overview

The DB4BM is a high power explosion proof, metallic loudspeaker, with outstanding intelligibility and acoustic performance.

Certified for use in a wide range of temperatures from -76°F to +185°F, the Ex enclosure is manufactured in either marine grade alloy or stainless steel with a rugged thermoplastic flare providing a corrosion free and aesthetically pleasing product.

The frequency response of the unit ensures that critical voice messages and general alarm tones are highly intelligible. The specific SPL figure for sensitivity, 1W @ 10 feet is 94dB whilst at 10 feet the 25W unit produces 109dB, the 15W unit 107dB and the 8W unit 104dB.

## Features

- UL certified for USA and Canada
  - Class I, Div 1, Groups A, B, C & D (CNL B, C & D)
  - Class I, Div 2, Groups A, B, C & D
  - Class I, Zone 1
  - Class II, Div 2, Groups F & G
  - Zone 21
  - Class III, Div. 2
  - Fire alarm and general use
- Certified temperature -60°C to +85°C (-76°F to + 185°F)
- IP66 & IP67 and NEMA 4x & 6
- SIL 2 certified
- Up to 109dB output @ 10 feet
- 8W, 15W & 25W versions
- Power tapings, via integral transformer
- Frequency response 350Hz - 8kHz
- Ex enclosure - Stainless steel or marine grade alloy
- Flare - High impact thermoplastic polyester
- Stainless steel mounting bracket and cover screws
- Mounting bracket has ratchet facility as standard
- Optional swivel bracket available
- Optional resettable fuse†

† Contact MEDC for details

Options include DC blocking capacitors for monitored systems, resettable fuses for compliance with marine regulations and a swivel bracket that gives the installer greater flexibility when positioning the unit. The short flare option is a worthy addition to the range offering a high SPL and wide dispersion angle in a compact unit.



| Certifications |                                                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UL Haz Locs    | UL certified for USA and Canada, listing no E203310                                                                                                                                                                                                                       |
| UL certified   | Class I, Div. 1 Groups A, B, C & D (CNL Groups B, C & D)<br>Class 1 Div. 2 Groups A, B, C & D<br>Zone 1 Ex db IIC Gb / AEx db IIC Gb T3/T4/T5/T6<br>Class II, Div. 2 Groups F&G<br>Zone 21 Ex tb IIIC Db / AEx tb IIIC Db T200°C/T135°C/T100°C/T85°C<br>Class III, Div. 2 |
| UL Ord Locs    | UL certified for USA and Canada, listing no S8447                                                                                                                                                                                                                         |
| SIL            | SIL2 certified to IEC61508. Cert number FSP 22008                                                                                                                                                                                                                         |

| Specifications                    |                                                                                                                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Material                          | Ex enclosure – marine grade alloy or stainless steel<br>Flare - flame retardant, high impact, UV stable, thermoplastic polyester (UV stability tested to ISO 4892 part 3)<br>Hardware - bracket, fixings and captive cover screws in 316 stainless steel |
| Fire retardancy                   | Outer flare - thermoplastic polyester. V0 flammability rating                                                                                                                                                                                            |
| Finish                            | Body - Painted Black<br>Flare - Natural black, natural red or painted as specified                                                                                                                                                                       |
| Rated power                       | 8W, 15W or 25W                                                                                                                                                                                                                                           |
| Frequency range                   | Frequency response 350Hz to 8kHz<br>8W and 15W units are available for use @ 200Hz                                                                                                                                                                       |
| Weight                            | Alloy 9¾ lbs/ 4.4 kg, stainless steel 18½ lbs/ 8.35 kg based on long flare DC unit                                                                                                                                                                       |
| Ingress protection                | IP66 & IP67* / NEMA 4x & 6<br>* Full IP rating applies to terminal chamber only. High pressure jets should not be aimed directly into the re-entrant horn. IP65 Short Flare version.                                                                     |
| Entries                           | Up to 2 x M20, M25, ½" or ¾" NPT<br>Blanking plug must be supplied with 2 x entries                                                                                                                                                                      |
| Terminals                         | 8 x 12AWG                                                                                                                                                                                                                                                |
| Mounting arrangement              | Stainless steel bracket with ratchet facility, optional swivel bracket available                                                                                                                                                                         |
| Labels                            | Optional duty and tag labels available                                                                                                                                                                                                                   |
| Dispersion angle                  | Long flare (short in brackets), -6dB, 1kHz = 140° (240°), 4kHz = 40° (50°)                                                                                                                                                                               |
| Sound pressure level at 10' (dB): |                                                                                                                                                                                                                                                          |

| Values in dB @ 10 feet<br>(Unsmoothed, peak values) | 8W    |      | 15W   |      | 25W   |      |
|-----------------------------------------------------|-------|------|-------|------|-------|------|
|                                                     | Short | Long | Short | Long | Short | Long |
| IIC Gas                                             | 101   | 104  | 104   | 107  | 106   | 109  |

Tolerance +/- 3dB

Certified temperature:

| Power   | Maxiumum Temperature                              |
|---------|---------------------------------------------------|
| 25 Watt | +85°C T3/T200°C, +70°C T4/T135°C, +40°C T5/T100°C |
| 15 Watt | +85°C T4/T135°C, +40°C T5/T100°C                  |
| 8 Watt  | +85°C T4/T135°C, +55°C T5/T100°C, +40°C T6/T85°C  |

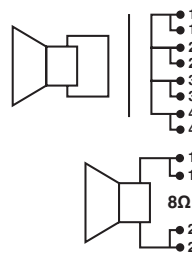
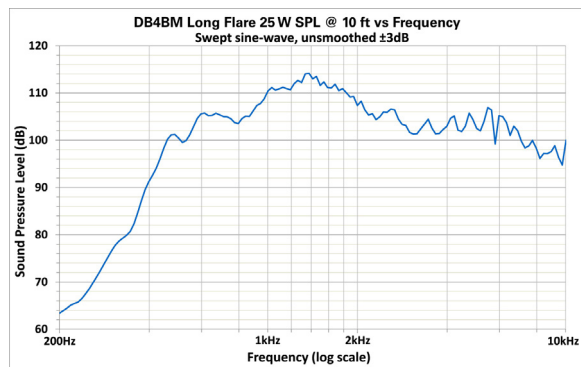
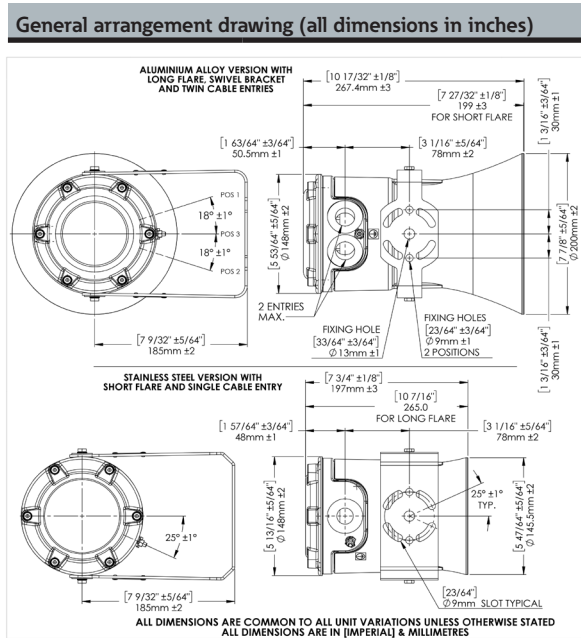
Transformer: Used by combining the rated power tappings below

| Tappings | Power |      |      |
|----------|-------|------|------|
|          | 8W    | 15W  | 25W  |
| 1:2      | 8.0   | 15.0 | 25.0 |
| 2:3      | 4.0   | 7.5  | 12.5 |
| 3:4      | 2.0   | 5.0  | 6.0  |
| 1:3      | 1.5   | 4.0  | 4.0  |
| 2:4      | 0.7   | 2.0  | 2.0  |
| 1:4      | 0.4   | 0.8  | 1.0  |

## Ordering requirements

The following code is designed to help in selection of the correct unit. Build up the reference number by inserting the code for each component into the appropriate box

| Model | Material                                                                                                | Certification                                                                                                                                                                                         | Wattage                                                                                                                              | Transformer                                                                                                                                                | Labels                                                                                                                    | Entries                                                                                                                                                                                                                                                                                                                                                                                                                  | Options                                                                                                                                                                        | Finish                                                                                                                                                                                                                                |
|-------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DB4BM |                                                                                                         |                                                                                                                                                                                                       |                                                                                                                                      |                                                                                                                                                            |                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                |                                                                                                                                                                                                                                       |
|       | <div>Finish</div> <div>Alloy</div> <div>Stainless Steel</div> <div>Code</div> <div>A</div> <div>S</div> | <div>Ex Certification</div> <div>UL</div> <div>UL (ordinary locations)</div> <div>Dual ATEX/UL</div> <div>Dual IECEx/UL</div> <div>Code</div> <div>UL</div> <div>UW</div> <div>AU</div> <div>JU</div> | <div>Wattage</div> <div>8 Watt</div> <div>15 Watt</div> <div>25 Watt</div> <div>Code</div> <div>8</div> <div>15</div> <div>25*</div> | <div>Transformer</div> <div>8Ω</div> <div>100V</div> <div>70V</div> <div>25V</div> <div>Code</div> <div>N</div> <div>100</div> <div>70</div> <div>25</div> | <div>Labels</div> <div>None</div> <div>Duty</div> <div>Tag</div> <div>Code</div> <div>N</div> <div>D*</div> <div>T*</div> | <div>Entries</div> <div>1 x M20</div> <div>2 x M20</div> <div>inc 1 x plug</div> <div>1 x ½" NPT</div> <div>2 x ½" NPT</div> <div>inc 1 x plug</div> <div>1 x M25</div> <div>2 x M25</div> <div>inc 1x plug</div> <div>1 x ¾" NPT</div> <div>2 x ¾" NPT</div> <div>inc 1x plug</div> <div>Code</div> <div>1B</div> <div>2BP</div> <div>1C</div> <div>2CP</div> <div>1D</div> <div>2DP</div> <div>1E</div> <div>2EP</div> | <div>Options</div> <div>None</div> <div>Swivel bracket</div> <div>Capacitor</div> <div>Short flare</div> <div>Code</div> <div>N</div> <div>B†</div> <div>C*</div> <div>S</div> | <div>Finish</div> <div>Natural black</div> <div>Natural red</div> <div>Yellow</div> <div>Grey</div> <div>Blue</div> <div>Special</div> <div>Code</div> <div>N</div> <div>R</div> <div>Y</div> <div>G</div> <div>B</div> <div>S*</div> |
|       |                                                                                                         | * Select 25 if 8Ω is required                                                                                                                                                                         |                                                                                                                                      | * Please specify                                                                                                                                           |                                                                                                                           | † Alloy only - stainless steel has swivel as standard<br>* Specify value                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                |                                                                                                                                                                                                                                       |



1) Loop in/loop out (4 x 2) power tap change; 8 terminals

2) Loop in/loop out (2 x 2) 8 Ω; 4 terminals